

Sentinel

A Fully Procedural Game Engine and Open-World Military Shooter

Nothing is loaded from a model, image or sound file: everything is generated from code and a seed.

Presented by

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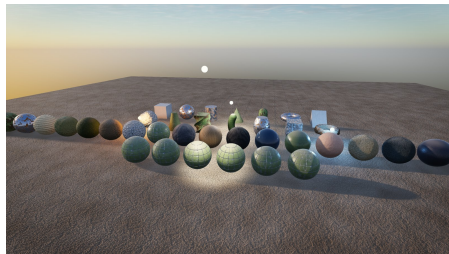
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1. Problem and objectives
2. Engine architecture
3. Procedural content: shapes and textures
4. The rendering pipeline
5. World, physics and gameplay
6. Results and verification
7. Challenges, limits and future work



Every material is a code-generated recipe.

1. Problem and objectives

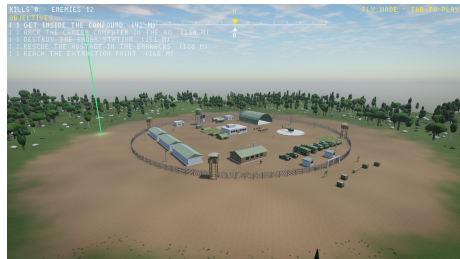
The question: can a complete, good-looking, playable 3D game be made by *code alone* on a student laptop?

Two hard limits

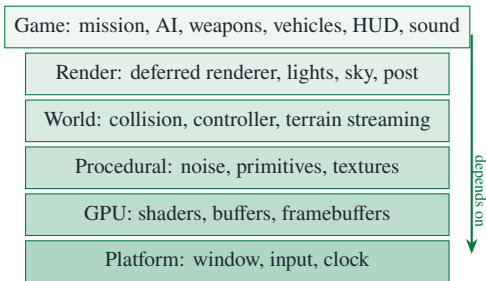
- No pre-saved model, image or precomputed data file (only shader text is read)
- macOS stops at OpenGL 4.1: no compute shaders

Objectives

1. Layered engine, dependencies only downward
2. All meshes, textures, sounds from code and seeds
3. Modern renderer: PBR, shadows, sky, post-processing
4. Playable game: vehicles, ladders, doors, mission
5. Verified by tests and a 16.7 ms frame budget



2. Engine architecture



A lower layer never knows a higher one.

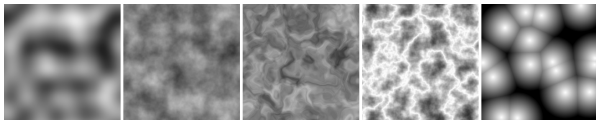
Readable by design

- `Main.odin` = table of contents: init, loop (input, update, render), shutdown
- One feature per file; procedures ≈ 40 lines
- Content is data: object = table of *part*, *transform*, *deformers*, *material*
- Every GL call checked; preconditions asserted

Size

164 Odin files ($\approx 19\,200$ lines), 47 GLSL files ($\approx 2\,000$ lines), 138 commits

3. Procedural content: shapes and textures



gradient, fractal, warped, ridged, cellular noise

- **Fractal noise** sums octaves, each $2\times$ finer and $\frac{1}{2}$ as strong; wrapped grid index $\Rightarrow$ **exact tiling**
- 18 recipes $\rightarrow$ 22 materials, baked once at 1024^2
- **Scharr** filter turns height into bumps
- Shapes: primitives + noise deformers; legs by two-bone IK

Fractal noise

$$F(\mathbf{x}) = \frac{\sum_{i=0}^{n-1} 2^{-i} v(2^i \mathbf{x})}{\sum_{i=0}^{n-1} 2^{-i}}$$

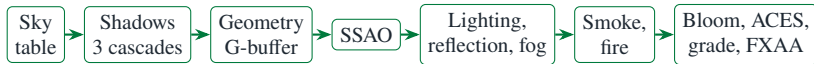
Normal from height

$$h_x \approx \frac{3(h_{NE} + h_{SE} - h_{NW} - h_{SW}) + 10(h_E - h_W)}{32\Delta N}$$
$$\hat{n} \propto (-kh_x, -kh_y, 1)$$

Vertex push

$$\mathbf{p}' = \mathbf{p} + A v(f\mathbf{p}) \hat{n}, \quad |\mathbf{p}' - \mathbf{p}| \leq A$$

4. The rendering pipeline



Deferred shading: first store each pixel's colour, normal, shininess, depth (the “data poster”), then light it once; many lights stay cheap.

Cook–Torrance reflection

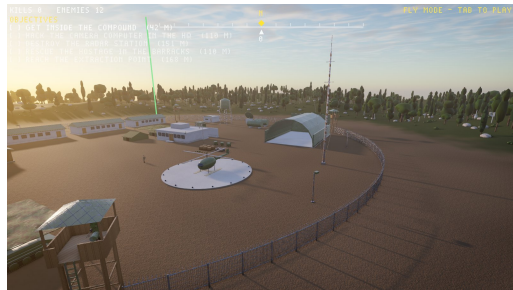
$$f = \frac{(1-F_I)(1-F_V)c_d}{\pi} + DVF$$

$$D = \frac{\alpha^2}{\pi((\mathbf{n} \cdot \mathbf{h})^2(\alpha^2 - 1) + 1)^2}, \quad F = F_0 + (1 - F_0)(1 - \mathbf{v} \cdot \mathbf{h})^5$$

Fog and tone map

$$C = TC_s + (1 - T)C_h, \quad T = e^{-kd}$$

$$C_{\text{out}} = \frac{x(2.51x+0.03)}{x(2.43x+0.59)+0.14}$$



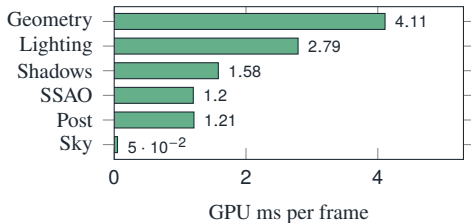
Dusk: sun, sky, haze, shadows, clouds

5. World, physics and gameplay

- **Sky:** sun height
 $\sin \theta = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos h$;
scattering sky; wind-driven cumulus and cirrus: $\mathbf{P}_{xz} = \frac{H}{d_y}(d_x, d_z) + \mathbf{w}t$
- **Terrain:** warped ridged noise, 64 m streamed chunks
- **Trees:** 4 species, per-tree size and tint, cantilever wind sway
- **Vehicles:** jeep, truck, carrier, tank, helicopter; bicycle steering $\dot{\psi} = \frac{v \tan \delta}{L}$, power fade $\dot{v} = a_0(1 - 0.6 \frac{|v|}{v_{\max}})$
- **Ladders:** 4-state grip machine, hands on rungs
- **Mission:** hack, destroy, rescue, extract; stealth AI; all sound synthesised



6. Results and verification



10.9 ms GPU of the 16.7 ms budget (1080p);
display locks 60 Hz.

Verification

Unit tests (13 packages)	281
TextureCheck (exact tiling)	8 318
RenderCheck	318 567
GpuCheck	30
Scripted playthroughs (5/5 objectives)	2



7. Challenge solved, limits and future work

Bug: a bright vertical line at night. Found by bisecting draw items, lights and passes. Cause: the fog colour sampled the *whole* sky, so one horizon star painted a streak down every surface in its column. Fix: fog uses the smooth atmosphere only.



before



after

Limitations

Blocky primitive models; no true global illumination; lumpy foliage; tuned by scripts and screenshots, not a big playtest.

Future work

Leaf cards, screen-space GI, point-light shadows, multiplayer, a compute-shader port.

Take-away

Most “art” is a few simple ideas (smooth noise, a lighting formula, transforms) applied consistently and tested.

- [1] K. Perlin, “Improving noise,” *ACM TOG (SIGGRAPH)*, 21(3), 2002.
- [2] S. Worley, “A cellular texture basis function,” *SIGGRAPH*, 1996.
- [3] R. Cook and K. Torrance, “A reflectance model for computer graphics,” *ACM TOG*, 1(1), 1982.
- [4] B. Walter et al., “Microfacet models for refraction through rough surfaces,” *EGSR*, 2007.
- [5] B. Karis, “Real shading in Unreal Engine 4,” *SIGGRAPH Course*, 2013.
- [6] S. Hillaire, “A scalable and production ready sky and atmosphere rendering technique,” *CGF*, 39(4), 2020.
- [7] R. Dimitrov, “Cascaded shadow maps,” *NVIDIA*, 2007.
- [8] M. Oren and S. Nayar, “Generalization of Lambert’s reflectance model,” *SIGGRAPH*, 1994.
- [9] K. Narkowicz, “ACES filmic tone mapping curve,” 2016.
- [10] J. Jimenez, “Next generation post processing in Call of Duty: Advanced Warfare,” *SIGGRAPH*, 2014.
- [11] T. Lottes, “FXAA,” *NVIDIA*, 2009.
- [12] M. Segal and K. Akeley, *The OpenGL Graphics System, v4.1*, Khronos, 2010.
- [13] I. Quilez, “Domain warping,” iquilezles.org.
- [14] S. Hargreaves and M. Harris, “Deferred shading,” *NVIDIA*, GDC 2004.
- [15] H. Scharr, “Optimal operators in digital image processing,” Ph.D., Heidelberg, 2000.
- [16] T. Akenine-Möller et al., *Real-Time Rendering*, 4th ed., CRC Press, 2018.

Thank you

Questions are welcome



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